



iTunes kids allowance shutting

Apple is shutting kids allowance for iTunes from May 25th and has already stopped taking new users.
<http://dgit.in/itunesallow>



FB unveiled bot for Messenger

Facebook has unveiled its messenger chat bot after Microsoft announced its bot for skype.
<http://dgit.in/FBvsMI>

Tomorrow's tech

be comparatively easier to spot. It is the third one that is the most dangerous and can even form internally. Although there are techniques to test for these without destroying the material, like ultrasound scanning, most of these defects happen during actual usage where such testing is not being done.

This is where the inspiration from our very own human body comes into play. The human body can actively sense damage, stop it from worsening and start working on fixing it in real time. It is this phenomenon that self-healing materials aim to replicate, without human intervention in the repair process.

How does it work?

Almost all the things that we see around us nowadays are made out of plastics, or as science likes to call it, polymers - smartphones, shoes, bottles, footballs, houses etc. So it is not unexpected that research into self-repairing materials is mainly focused on polymers. This research has been mainly ongoing since early 2000s and one of the first breakthroughs was achieved by the team of Scott White, Nancy Sottos, and their colleagues from the University of Illinois at Urbana Champaign.

Embedded Healing Agents

The most approachable self-healing technology involves micro-capsules of adhesive substance embedded into materials. When the material cracks or is under stress, these capsules break open and the adhesive material, or glue, comes out to fill the cracks and solidify. This glue can be simply a strong adhesive that holds the material together. In another approach, where the material itself is mainly a solid polymer, this glue-like material is a liquid made of the corresponding monomer (the basic constituent of the polymer). Now, when the material cracks, the capsules break open, and the polymer is exposed to the monomer resulting in further polymerization that creates more of the original polymer which fills the damaged area. Typically, this process might need a powdered chemical catalyst to be included for it to successfully occur at normal temperature and conditions without intervention.

The main problem with this approach is that the embedding might structurally weaken the original material. Due to that reason, there is a size restriction on the capsules. Also, this is a one-time method, because once the capsules are cracked, another instance of damage at the same location will have nothing to rely on. And that second instance is highly likely due to the weakening of the material after repair.

Also, since microcapsules have to be below a certain size threshold, there is a very strict limit to the size of cracks that can be filled.

Microvascular Healing

Taking the human body analogy one step further, microvascular healing technology implements micro-tube like structures within the material that connect to a pressurised source of adhesive material. Wherever a crack occurs in the material, the tube is depressurized at that end and the material flows into the crack. This method can seal cracks up to ten times the size that can be fixed by micro-capsules.

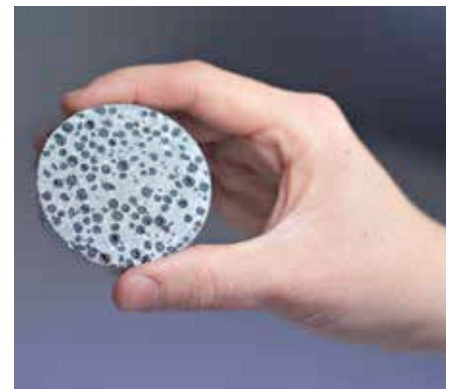
This method has its own drawbacks as well, since some cracks will spread faster than they can be repaired. But on long term cracks, like those that appear on pavements and buildings, this method is certainly useful.

Memory Materials

There is a particular classification of materials that has a tendency to remember and retain its original shape. Don't worry, there is no actual memory involved here and your favorite shoe is not suddenly

going to become sentient. Memory materials are all around us, like the frames of the eyeglasses that we wear, which snap back to their original shape even if we bend them.

Some of these materials require an external influence like heat or light to repair themselves. This can be achieved by a network of fibre-optic cables similar to the microvascular network described in the previous method. But this network



Self Repairable concrete can extend the durability of construction significantly!

would bring heat and light to the point of damage, causing the material to alter back to its original configuration. In fact, it has the added advantage of making the material fibre-reinforced.

So what's really out there already?

Although it is true that full scale self repair technology is not even present in the labs as of now, there have been a lot of interesting developments in that direction that show a lot of promise.

When Hendrik Jonkers, Microbiologist at Delft University, was looking for a way to strengthen concrete, he didn't turn towards the conventional methods of steel or stone reinforcement. Rather, he took inspiration from his own domain of biology. His solution involves using limestone-producing bacteria in the mixture for concrete, which can close up the gaps eventually. While the concrete is being mixed, specially selected types of bacteria along with nitrogen, phosphorus and calcium lactate are added as ingredients. These constituents can stay dormant in the concrete for upto 200 years. When-

SOMETIMES, ALL THAT YOU NEED IS A LITTLE HEAT

Thermosetting polymers break down into their original monomers if they are heated. When allowed to cool, these monomers join back into the polymer. Although this heat might seem like a broad assumption, in some situations it is readily available. Imagine an Armored Vehicle with the glasses made with such a material. The impact of a bullet would generate enough heat for the material to heal itself.